

LUMINARY ACADEMY

Mathematics Workbook

Grade 10 Mathematics

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Chapter 1

Quadratic Equations & Functions

Learning Objectives

- Solve quadratic equations by factoring, completing the square, and the quadratic formula
- Graph parabolas and identify vertex, axis of symmetry, and intercepts
- Determine the nature of roots using the discriminant
- Solve real-world problems modelled by quadratic equations

Key Formulas

- Standard form: $ax^2 + bx + c = 0$
- Quadratic Formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- Discriminant: $\Delta = b^2 - 4ac$
- $\Delta > 0$: two distinct real roots
- $\Delta = 0$: one repeated real root
- $\Delta < 0$: no real roots (complex)
- Vertex: $(-b/2a, f(-b/2a))$
- Axis of symmetry: $x = -b/2a$
- Completing the square: $x^2 + bx = (x + b/2)^2 - (b/2)^2$

1.1 Worked Examples

Example 1

Solve by factoring: $x^2 - 5x + 6 = 0$

Solution:

$$(x - 2)(x - 3) = 0 \quad x = 2 \text{ or } x = 3$$

Example 2Solve using the quadratic formula: $2x^2 - 3x - 2 = 0$ **Solution:**

$$a = 2, b = -3, c = -2 \quad \Delta = (-3)^2 - 4(2)(-2) = 9 + 16 = 25 \quad x = (3 \pm 5) / 4 \quad x = 2 \text{ or } x = -1/2$$

Example 3Find the vertex and axis of symmetry of $y = 2x^2 - 8x + 3$ **Solution:**

$$\text{Axis: } x = -(-8)/(2 \times 2) = 8/4 = 2 \quad \text{Vertex } y: 2(4) - 8(2) + 3 = 8 - 16 + 3 = -5 \quad \text{Vertex: } (2, -5), \\ \text{Axis: } x = 2$$

Example 4A ball is thrown upward. Height $h = -5t^2 + 20t + 2$. Find max height and time to reach it.**Solution:**

$$\text{Max at } t = -20/(-10) = 2 \text{ seconds } h(2) = -5(4) + 40 + 2 = -20 + 42 = 22 \text{ metres}$$

Common Mistakes

- When using the quadratic formula, compute the discriminant first to check the type of roots.
- The vertex x-coordinate is $-b/2a$, not $b/2a$ — the negative sign matters!

1.2 Practice**Independent Practice**1. Solve by factoring: $x^2 + 3x - 10 = 0$

Answer: _____

2. Solve using quadratic formula: $x^2 - 4x + 1 = 0$

Answer: _____

3. Solve by completing the square: $x^2 + 6x - 7 = 0$

Answer: _____

4. Find the discriminant and describe roots: $3x^2 - 2x + 5 = 0$

Answer: _____

5. Sketch $y = x^2 - 4x + 3$ (label vertex, axis, x-intercepts)

Answer: _____

6. A rectangular garden has area 60 m^2 . Length is 4 m more than width. Find dimensions.

Answer: _____

7. Solve: $(x + 1)^2 = 16$

Answer: _____

8. Find the value of k for which $x^2 + kx + 9 = 0$ has equal roots.

Answer: _____

1.3 Exit Ticket

Exit Ticket

1. What is the discriminant of $2x^2 + 3x + 1 = 0$?

Answer: _____

2. Solve: $x^2 - 9 = 0$

Answer: _____

3. What is the axis of symmetry of $y = x^2 - 6x + 5$?

Answer: _____

4. Does $2x^2 + x + 3 = 0$ have real solutions? Why?

Answer: _____

5. Expand and simplify: $(x + 3)^2 - 4$

Answer: _____

Chapter 2

Simultaneous Equations

Learning Objectives

- Solve linear simultaneous equations by substitution and elimination
- Solve one linear and one quadratic equation simultaneously
- Interpret simultaneous equations graphically
- Apply simultaneous equations to real-world problems

Methods

- Substitution: solve one equation for a variable, substitute into the other
- Elimination: multiply equations to match coefficients, then add/subtract
- Graphical: find intersection point(s) of two graphs
- Linear + quadratic: substitute the linear equation into the quadratic

2.1 Worked Examples

Example 1

Solve by elimination: $3x + 2y = 12$ and $5x - 2y = 4$

Solution:

Add equations: $8x = 16 \rightarrow x = 2$ **Substitute:** $3(2) + 2y = 12 \rightarrow 2y = 6 \rightarrow y = 3$ **Solution:** $(2, 3)$

Example 2

Solve by substitution: $y = 2x - 1$ and $3x + y = 9$

Solution:

$$3x + (2x - 1) = 9 \quad 5x = 10 \quad x = 2, y = 2(2) - 1 = 3 \quad \text{Solution: } (2, 3)$$

Example 3

Solve simultaneously: $y = x + 1$ and $y = x^2 - 1$

Solution:

$$x + 1 = x^2 - 1 \quad x^2 - x - 2 = 0 \quad (x - 2)(x + 1) = 0 \quad x = 2: y = 3 \mid x = -1: y = 0 \quad \text{Solutions: } (2, 3) \text{ and } (-1, 0)$$

2.2 Practice

Independent Practice

1. Solve: $2x + y = 7$ and $x - y = 2$

Answer: _____

2. Solve: $4x - 3y = 1$ and $2x + y = 5$

Answer: _____

3. Solve: $x + 2y = 8$ and $3x - y = 3$

Answer: _____

4. Solve: $y = x - 3$ and $x^2 + y^2 = 25$

Answer: _____

5. Two numbers sum to 30. Their difference is 6. Find them.

Answer: _____

6. Solve: $y = 3x$ and $y = x^2 + 2x$

Answer: _____

7. Solve: $2x - y = 4$ and $y = x^2 - 2$

Answer: _____

8. A cinema sells adult tickets at \$12 and child tickets at \$7. 100 tickets sold, total \$910. How many of each?

Answer: _____

Chapter 3

Geometry — Circles & Polygons**Learning Objectives**

- Apply circle theorems: angles in semicircle, tangent-radius, chord bisector
- Calculate arc length, sector area, and segment area
- Find interior and exterior angles of polygons
- Use properties of similar and congruent figures

Key Formulas

- Circumference: $C = 2\pi r$
- Circle Area: $A = \pi r^2$
- Arc Length: $L = (\theta/360^\circ) \times 2\pi r$ (θ in degrees)
- Sector Area: $A = (\theta/360^\circ) \times \pi r^2$
- Sum of interior angles of n-gon: $(n-2) \times 180^\circ$
- Each interior angle of regular n-gon: $(n-2) \times 180^\circ / n$
- Each exterior angle of regular n-gon: $360^\circ / n$

3.1 Worked Examples**Example 1**

A sector has radius 6 cm and central angle 120° . Find arc length and area.

Solution:

$$\text{Arc} = (120/360) \times 2\pi(6) = (1/3)(12\pi) = 4\pi \approx 12.6 \text{ cm} \quad \text{Area} = (120/360) \times \pi(36) = 12\pi \approx 37.7 \text{ cm}^2$$

Example 2

Find the sum of interior angles of a heptagon (7 sides). Find each angle if regular.

Solution:

$$\text{Sum} = (7-2) \times 180^\circ = 5 \times 180^\circ = 900^\circ \text{ Each angle} = 900^\circ / 7 \approx 128.6^\circ$$

Example 3

An angle in a semicircle: If AB is a diameter and C is on the circle, what is angle ACB?

Solution:

$$\text{By the theorem: angle in a semicircle} = 90^\circ \text{ Angle ACB} = 90^\circ$$

3.2 Practice

Independent Practice

1. A circle has diameter 14 cm. Find circumference and area.

Answer: _____

2. Find arc length: radius 10, central angle 72° .

Answer: _____

3. Find sector area: radius 8, central angle 45° .

Answer: _____

4. Find interior angle sum of a dodecagon (12 sides).

Answer: _____

5. A regular pentagon — find each interior and exterior angle.

Answer: _____

6. Two tangents from an external point are always equal in length — prove conceptually.

Answer: _____

7. A chord is 16 cm long. The radius is 10 cm. Find the distance from the chord to the centre.

Answer: _____

8. Find the missing angle: an inscribed angle subtends an arc of 140° .

Answer: _____

Chapter 4

Trigonometry — Right Triangles

Learning Objectives

- Define sine, cosine, and tangent ratios in right triangles
- Find unknown sides and angles using trigonometric ratios
- Apply the Pythagorean theorem
- Solve problems using angles of elevation and depression

Key Formulas — SOH CAH TOA

- $\sin \theta = \text{Opposite} / \text{Hypotenuse}$
- $\cos \theta = \text{Adjacent} / \text{Hypotenuse}$
- $\tan \theta = \text{Opposite} / \text{Adjacent}$
- Pythagorean Theorem: $a^2 + b^2 = c^2$ (c = hypotenuse)
- Special angles: $\sin 30^\circ = 0.5$, $\cos 30^\circ = \sqrt{3}/2$, $\tan 30^\circ = 1/\sqrt{3}$
- Special angles: $\sin 45^\circ = \sqrt{2}/2$, $\cos 45^\circ = \sqrt{2}/2$, $\tan 45^\circ = 1$
- Special angles: $\sin 60^\circ = \sqrt{3}/2$, $\cos 60^\circ = 0.5$, $\tan 60^\circ = \sqrt{3}$

4.1 Worked Examples

Example 1

In right triangle ABC, angle B = 90° , angle A = 35° , AB = 8 cm. Find BC and AC.

Solution:

$$\tan 35^\circ = \text{BC}/\text{AB} \rightarrow \text{BC} = 8 \times \tan 35^\circ \approx 8 \times 0.700 \approx 5.60 \text{ cm}$$
$$\cos 35^\circ = \text{AB}/\text{AC} \rightarrow \text{AC} = 8/\cos 35^\circ \approx 8/0.819 \approx 9.77 \text{ cm}$$

Example 2

A ladder 10 m long leans against a wall. It makes a 65° angle with the ground. How high up the wall does it reach?

Solution:

$$\sin 65^\circ = \text{height}/10 \quad \text{height} = 10 \times \sin 65^\circ \approx 10 \times 0.906 \approx 9.06 \text{ m}$$

Example 3

From the top of a 40 m cliff, the angle of depression to a boat is 28° . Find the horizontal distance to the boat.

Solution:

$$\tan 28^\circ = 40 / d \quad d = 40 / \tan 28^\circ \approx 40 / 0.532 \approx 75.2 \text{ m}$$

4.2 Practice

Independent Practice

1. In right triangle: hypotenuse = 13, one leg = 5. Find the other leg.

Answer: _____

2. Find sin, cos, tan of 30° using the exact values (no calculator).

Answer: _____

3. Angle of elevation to top of a tree is 40° , standing 15 m away. Find height.

Answer: _____

4. Find angle θ : opposite = 7, adjacent = 24.

Answer: _____

5. A ramp rises 2 m over a horizontal distance of 8 m. Find the angle of inclination.

Answer: _____

6. In right triangle: angle = 50° , hypotenuse = 20. Find both legs.

Answer: _____

7. Two buildings are 30 m apart. From the top of the shorter (height 25 m), the angle of elevation to the top of the taller is 32° . Find the taller building's height.

Answer: _____

8. Verify: $\sin^2 45^\circ + \cos^2 45^\circ = 1$

Answer: _____

Chapter 5

Probability & Counting Principles

Learning Objectives

- Calculate theoretical and experimental probability
- Apply addition and multiplication rules for probability
- Use permutations and combinations for counting
- Construct and interpret probability trees and tables

Key Formulas

- $P(A) = \text{favourable outcomes} / \text{total outcomes}$
- $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$ [Addition Rule]
- $P(A \text{ and } B) = P(A) \times P(B|A)$ [Multiplication Rule]
- Independent events: $P(A \text{ and } B) = P(A) \times P(B)$
- Complement: $P(A') = 1 - P(A)$
- Permutation: $nPr = n! / (n-r)!$
- Combination: $nCr = n! / (r!(n-r)!)$

5.1 Worked Examples

Example 1

A bag has 4 red, 3 blue, 5 green balls. One drawn at random. Find $P(\text{red})$, $P(\text{not green})$, $P(\text{red or blue})$.

Solution:

Total = 12 $P(\text{red}) = 4/12 = 1/3$ $P(\text{not green}) = 7/12$ $P(\text{red or blue}) = 7/12$

Example 2

How many ways can 5 students be arranged in a line?

Solution:

$${}^5P_5 = 5! = 5 \times 4 \times 3 \times 2 \times 1 = 120 \text{ ways}$$

Example 3

A committee of 3 is chosen from 8 people. How many ways?

Solution:

$${}^8C_3 = 8!/(3! \times 5!) = (8 \times 7 \times 6)/(3 \times 2 \times 1) = 56 \text{ ways}$$

Example 4

Two dice rolled. Find $P(\text{sum} = 7)$.

Solution:

$$\text{Favourable: } (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) = 6 \text{ outcomes Total} = 36 \quad P(\text{sum} = 7) = 6/36 = 1/6$$

5.2 Practice

Independent Practice

1. A card is drawn from a standard deck (52). Find $P(\text{ace})$, $P(\text{heart})$, $P(\text{red face card})$.

Answer: _____

2. Two coins flipped. Find $P(\text{both heads})$, $P(\text{at least one tail})$.

Answer: _____

3. In how many ways can the letters of MATH be arranged?

Answer: _____

4. From 10 students, 4 are selected for a team. How many combinations?

Answer: _____

5. $P(A) = 0.4$, $P(B) = 0.3$, $P(A \text{ and } B) = 0.12$. Find $P(A \text{ or } B)$.

Answer: _____

6. Bag has 6 red, 4 blue. Two drawn without replacement. $P(\text{both red})$?

Answer: _____

7. A 4-digit PIN uses digits 0-9, no repetition. How many possible PINs?

Answer: _____

8. $P(\text{rain today}) = 0.6$. $P(\text{no rain today and no rain tomorrow}) = 0.12$. Find $P(\text{rain tomorrow})$.

Answer: _____

5.3 Exit Ticket

Exit Ticket

1. $P(A) = 0.7$. What is $P(A')$?

Answer: _____

2. How many ways to choose 2 items from 5?

Answer: _____

3. Are rolling a 6 and flipping heads independent events?

Answer: _____

4. A bag has 3 red, 2 blue. $P(\text{blue})$ on first draw?

Answer: _____

5. What is the difference between a permutation and a combination?

Answer: _____

Answer Key — Selected Problems

Ch.1 #1: $x^2 + 3x - 10 = (x+5)(x-2) = 0 \rightarrow x = -5 \text{ or } x = 2$

Ch.1 #4: $\Delta = 4 - 60 = -56 < 0$: no real roots

Ch.1 #8: Equal roots when $\Delta = 0$: $k^2 - 36 = 0 \rightarrow k = \pm 6$

Ch.2 #1: $x = 3, y = 1$

Ch.2 #5: Numbers are 18 and 12

Ch.3 #1: $C = 14\pi \approx 44.0 \text{ cm}$; $A = 49\pi \approx 153.9 \text{ cm}^2$

Ch.3 #5: Interior = 108° , Exterior = 72°

Ch.4 #1: leg = $\sqrt{(169-25)} = \sqrt{144} = 12$

Ch.4 #5: $\theta = \arctan(2/8) = \arctan(0.25) \approx 14.0^\circ$

Ch.5 #1: $P(\text{ace})=1/13$, $P(\text{heart})=1/4$, $P(\text{red face})=6/52=3/26$

Ch.5 #7: $10 \times 9 \times 8 \times 7 = 5040 \text{ PINs}$